

Two data sets: creativity and tomato

Presentation Outline

- 1 Data sets
 - The creativity study
 - The tomato yield study
- 2 Where is randomization in these studies?
- 3 Important points

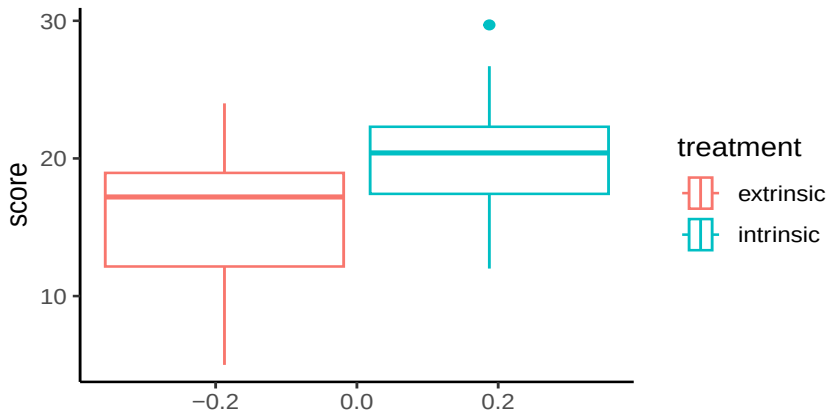
The creativity study

The creativity study: Sleuth Chapter 1, Case study 1

- Context: Behavioral science, comparison of two motivational schemes
 - Intrinsic motivation:
do something because of curiosity, pleasure, challenge of task or personal satisfaction
 - Extrinsic motivation:
do something because of external rewards or avoiding negative consequences
- Question: Does motivation scheme influence creativity?
- Design: "Subjects with considerable experience in creative writing were randomly assigned to one of two treatment groups."
- Response: Creativity score on a poem each individual wrote
- Details of the study and reference to the paper: Chapter 1, Case study 1 in the Sleuth

The data

23 extrinsic, 24 intrinsic individuals



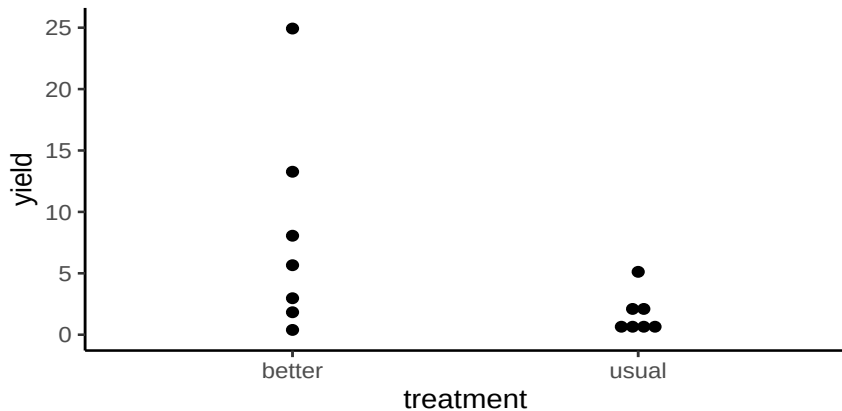
The tomato yield study

The tomato yield study

- Context: Backyard garden: comparison of 2 fertilizers
 - Usual: The fertilizer I usually used
 - Better: More expensive fertilizer, advertised to increase yield
- Question: Does type of fertilizer affect yield?
- Design: Plants purchased from a local garden store, planted in my garden, then randomly assigned to usual or better fertilizer
- Response: Yield of tomatoes, total over the season

The data

7 usual, 7 better plants



Where is randomization in the creativity study?

- Reminder: 2 possible sources of randomization
 - Random choice of units
 - Random assignment of treatments
- What sort(s) of randomization were used in the creativity study?
 - Choice of units?
 - Assignment of treatments?
- What is best name to describe this study: Survey, Observational, Experiment?

Where is randomization in the tomato study?

- What sort(s) of randomization were used in the tomato study?
 - Choice of units?
 - Assignment of treatments?
- What is best name to describe this study: Survey, Observational, Experiment?

Important points:

Important points:

- Two studies
 - Random assignment of treatments: Yes for both
 - Random choice of units: No for both
Convenience sample for both studies
 - Both should be called experiments
 - Or randomized experiments